Branch norm: independent source review

5 November 2026

Scoped source fidelity supported; machine unverified. Published revision `afccdb7d86ba`; independent derivation `c1dedb6021d1` was committed before candidate inspection.

Three constructions. Actual branch stalk and geometric DVR instances are used. A numerator root gives an $x - a$ factor; a nonzero scalar is a unit. Existing order-one lemmas at B and $-B$ give conjugate orders one and one; additive valuation gives product order two. Finiteness follows from those explicit orders. Norm evaluation gives the numerator root. Differentiating the norm at two support roots gives $b^2 f'(a) = 0$, hence $b = 0$ at a simple branch.

Fixture checked by hand. Over $\mathbf{Q}$, $f = x^5 - x$, $u_1 = x(x-1)$, $u_2 = x^2 + 1$, coefficient tuple $(-2, 0, 0, -4, -4)$, $n = 0$ and $U = -4(x+1)$ satisfy $-u_1 u_2 U = 4f$. Base orders at zero are $(1, 0, 0)$; stalk orders are $(2, 0, 0)$.

Exact boundary. Mapped curve/norm/support identities and base-multiplicity transport remain absent, as disclosed. The new theorem concerns the explicit conjugate product, not yet the support-factor sum. The local statements hold over any field with simple branch; in characteristic two the conjugates coincide, so no distinct-cover interpretation follows.

Checks. Manifest and exact Git bytes match. Existing primary APIs support the proof structure. No compiler, named axiom audit or execution occurred. Producer lease expiry is coordination history, not mathematical evidence.